Answers and talk notes

Ratio Reasoning: Convert Measurements Between Systems

What your student is learning

Tonight we solve problems that need TWO steps in order: convert to the unit the rate uses, and only then apply the rate.

Answers

Ask your student to explain before you look.

1. **D.** 60. 20 × 3 = 60, so the dose is 60 mg.
2. **A.** 20. 44 ÷ 2.2 = 20, so 44 pounds is about 20 kilograms.
3. **B.** Change 55 pounds to kilograms. Change 55 pounds to kilograms first, because the rate is per kilogram.
4. **C.** 15. 33 ÷ 2.2 = 15, so the dog weighs about 15 kilograms.
5. **A.** 50. 25 × 2 = 50, so the dose is 50 mg.

Talk about it

Why must you change the unit before you use the rate?

Home mission

Find a bag of rice or flour labeled in pounds. Pretend a made-up rate: 2 mg for every 1 kilogram. Ask your student which step comes first. Real doses always come from a vet or doctor.

When they are stuck, ask

- What unit does the rate use? Are we in that unit yet?
- Which step has to come first, and why?
- What unit does your final answer have?

Try not to say

- Multiply by the rate first, convert at the end.
- 5 kilometers and 5 miles are about the same thing.
- Leave off the unit, everyone knows what it is.